Functions & Proportionality

Lesson 1 / Overview

Students will explore concrete examples of linear and quadratic functions. They will build tables, and explain, using informal language, patterns observed. Function rules will be recorded algebraically. For linear functions, two rules will be recorded.

Materials
POD: Transparency of POD 1
Student Pages:
Number Patterns from Cutting String; Patterns with Squares; Number Patterns Equil Triangles; Pract Building Tables
Other Materials:
string, scissors, pattern blocks; Transparency of course goals; transpwrency of Summary Tables

Teaching Actions

POD: What are the last two digits of this number written in standard form: 7^{9999}?

Patterns observed:

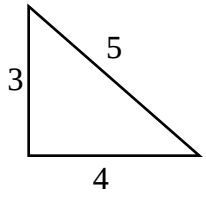
$7^1 = 07$	$7^5 = 07$
$7^2 = 49$	$7^6 = 49$
$7^3 = 43$	$7^7 = 43$
$7^4 = 01$	$7^8 = 01$

One possible solution: $9999 \div 4 = 2499$ remainder 3. All exponents with the same remainder have the same last two digits. Expect a variety of solutions.

Large Group Introduction

1. Start class with a discussion of the role patterns play in mathematics. Much of what mathematicians do is to look for regularities in real and mathematical worlds and often use symbols to describe generalizations that can be made.

Example: 1000 years before Pythagoras, the Babylonians observed the relationship among sides of a right triangle.



$a^2 + b^2 = c^2$

2. Explain that today and the rest of the course they will be exploring patterns and regularities from different explorations and that they will be finding different ways to record their observations.

Comments

1. After completing the POD, share with students the class goals using transparency of goal sheet found in teacher's guide. Students then complete their self reflection on course goals before doing the first large group activity. This form also is found in the teacher's guide.

2. Each class starts with a **Problem of a Day**. This is to engage students as they enter class.

Each class ends with students completing in their groups the **Big Idea** form. They are to write what they consider to be the important ideas developed in that day's lesson.

After Day 1 the **Big Idea** forms are used to review mathematics developed from the precious day. After discussing the POD, the instructor reads the forms and adds to them as needed to highlight the important ideas and connections made the previous day.

Each class then proceeds with the new topic for the day. This is done with large group introduction of the exploration that is completed during small group work. Large group discussion ends that sequence. This sequence may be repeated after the break.

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Teaching Actions

3. Set the stage for the string activity. Take a piece of string and cut it. Observe how many pieces there are. Ask: If I do that again, but cut twice, how many pieces of string will I have? Cut and verify guess. Complete a table and describe several patterns in the table.

Ask for patterns that could be used to extend the table to 10, 15, 100 cuts. Ask for the relationship to find the # of pieces given the # of cuts; the # of cuts given the number of pieces. **[p = c + 1; c = p - 1]**

# of cuts	0	1	2	3	4	5	6
# of pieces	1	2	3	4	5		

Small Group Work

5. Have students complete **Number Patterns from Cutting String**. As students work, observe the types of patterns that they describe. If they are only recording the function rule, ask them to describe at least three different patterns observed in the table.

Large Group Discussion

6. Share patterns observed from conditions 1, 2 and 3. Don't limit the discussion to just the function rules. Record function rules algebraically connecting the use of variable with their informal description of the rules.

For example: Condition 1: "To find the number of pieces you double the number of cuts and add one".
[p = 2c + 1]

Consider the second rule for each table.

For example, for condition 1, write the rule predicting the number of cuts given the number of pieces:

If the **number of pieces equals two times the number of cuts plus one**, then the **number of cuts equals the number of pieces minus one divided by two**.
[c = (p - 1)÷ 2].

Ask how the two rules are related.

Comments

3. Students should observe as many patterns as possible using their informal language prior to recording patterns algebraically.

While the function rules are the patterns you will highlight, it is important to show how other patterns can be used to extend the table.

# of cuts	0	1	2	3	4	5	6
# of pieces	1	2	3	4	5		

Students may note that if the # of cuts is even, the # of pieces is always odd; that the number of pieces is increasing by one as the # of cuts increases by one.

4. Discuss which patterns can be used to extend the table. Talk about how the function rule is the most efficient one to use when extending the data to 100 cuts.

5. Students often need clarification on how to "loop" the string around the scissors for page 2 of the string activity.

6. The function rules are:

Cond # 1: $p = 2c + 1$; $c = (p-1)/2$
Cond# 2: $p = L + 2$; $L = p - 2$
Cond # 3: $p = 2L + 2$; $L = (p-2)/2$
Extensions: $p = 2L + 3$; $p = 3L + 4$; $p = 4L + 5$

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Teaching Actions

Break

Small Group Work

7. Continue the exploration of patterns by completing in groups the following two activity sheets; **Patterns with Squares** and **Number Patterns from Trains of Equilateral Triangles.**

8. The function rule for **Patterns with Squares** is as follows:

Area = sum of the numbers 1 to t (t= the train number)

Note: This rule will be difficult to record algebraically; verbal explanation of what to do to find the answer is enough.

Perimeter = 4t (t = train number) ; Train # = P/4

10. **Number Patterns from Trains for Equilateral Triangles**

Students may need to be reminded that the unit of area for both parts of this activity is the green triangle.

Area = triangle number squared ($A = t^2$); (t = sq root of A)

Extension problem: Area = two times the rhombus number squared. **($A = 2\ r^2$; r = rhombus number).**

Large Group Discussion

11. Look for constant differences in all examples examined today. Use a transparency of the Summary Tables page. Sort problems into two groups by looking for constant differences in the tables. Match examples in each group with their functions. Note the differences in rules. Ask: Which group includes rules with only simple arithmetic? Which group includes rules with an exponent of two?

12. **Practice Building Tables** is provided if needed. Here students build tables given the rules and guess the rules, given the tables.

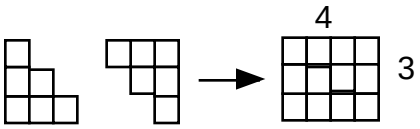
13. End class by completing the Big Idea form.

Comments

6. Students encounter examples of quadratic rules in the next two activities.

Patterns with Squares:
Area formula reflects the rule: $y = n(n+1)\div 2$. This is Gauss' rule for adding consecutive numbers. You will need to provide this rule.

You may want to model this rule using tiles:



sum of numbers 1 to 3 = 3(3+1)/2

For this example it is important to look at patterns across the table and compare those patterns with ones in other tables.

train #	1	2	3	4	5
area	1	3	6	10	15
		2	3	4	5
			1	1	1

7. For quadratic rules, it takes two differences to reach a constant. For linear rules, a constant difference occurs the first time.

train #	1	2	3	4	5
perim	4	8	12	16	20
		4	4	4	4

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Summary Tables

# of loops	# of pieces
1	
2	
3	
4	
5	

train #	Perimeter
1	
2	
3	
4	
5	

train #	Area
1	
2	
3	
4	
5	

triangle #	Area
1	
2	
3	
4	
5	

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Number Patterns from Cutting String

Continue exploring number patterns from cutting string as described in each condition below:

CONDITION 1: Fold the string in half. Make 1 cut. How many pieces of string do you have? Continue with another piece of string folded in half, making 2, 3 , 4 and 5 cuts. Complete the table below.

# of cuts	0	1	2	3	4	5
# of pieces						

QUESTIONS ??????????

1. Describe patterns that you observe in the table.

2. Without cutting the string, use a pattern from the table to determine the number of pieces for 7 cuts. How many cuts would yield 21 pieces of string?

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CONDITION 2: Loop the string through the scissors as shown, then cut. Find how many pieces you get for each number of loops. Complete the table below:

# of loops	0	1	2	3	4	5
# of pieces						

QUESTIONS ??????????

1. Describe patterns that you observe in the table.

2. Without cutting the string, use a pattern from the table to determine the number of pieces for 7 loops. How many loops would yield 12 pieces of string?

CONDITION 3: Tie ends of string together and then loop it through the scissors and cut. How many pieces do you get for each number of loops?

# of loops	0	1	2	3	4	5
# of pieces						

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Questions????????????

1. Describe patterns:

2. If the number of loops is 11, then how many pieces?
   If the number of pieces is 26, then how many loops?

3. If the number of loops is 23, then how many pieces?
   If the number of pieces is 52, then how many loops?
   Describe how you considered these two last questions.

Extensions: How many pieces will you get if the string is folded in each of these ways before being looped around the scissors? Guess and answer for 0, 1, 2 and 3 loops. Verify by actually cutting string. Predict for "n" loops.

a)

b)

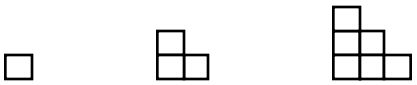
c)

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Patterns with Squares: Area and Perimeter

Using the orange pattern blocks, build the first three trains shown below:

train #



1 2 3

Build train numbers 4, 5, and 6. Draw a picture of each above.

Find area and perimeter of each train and record the data below:

Train #	Perimeter
1	
2	
3	
4	
5	
6	

Train #	Area
1	
2	
3	
4	
5	
6	

Describe patterns observed for area and perimeter.

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Questions ??????

1. What would the perimeter be for train 20? the nth train? Describe your strategy for finding out.

2. What would the area be for train 8? Describe your strategy for finding out.

3. Finding area for the 20th train or nth train is challenging. Play around with this and then describe your thoughts.

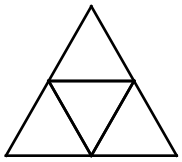
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Number Patterns from Trains of Equilateral Triangles

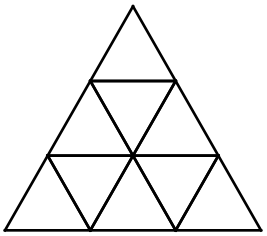
Using green pattern blocks form the first three triangles shown below. Notice that the triangles formed from the equilateral green triangles are also equilateral.



1st



2nd



3rd

Construct the fourth triangle in this series. If the unit of area is 1 green triangle, what is the area of the other triangles built? Record in the table below:

triangle number	1	2	3	4	5	6
area in green triangles						

Describe patterns that you observe in the table:

Predict the area of triangles 5 and 6. Verify by building them.

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It is possible to predict the area of the triangle given the triangle number. Describe in words how you could determine the area of the 20th triangle in the series.

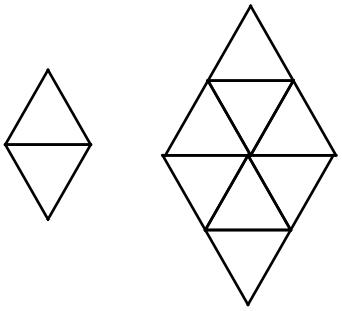
Translate the rule to an algebraic equation for the nth triangle in the series.

What if you knew the area is 441 green triangles. What triangular number would this be matched with?

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EXTENSION:

Instead of building triangles, build rhombuses with green triangles as shown below. Again let the **unit of area = 1 green triangle.** Complete the table.



rhombus number	1	2	3	4	5	6
area in green triangles						

Describe patterns observed in the table.

What is the area of rhombus 9? Rhombus 12? Rhombus N?

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Practice Building Tables

I. Build a table of 5 sample points that fit each rule:

double the #, subtract one

x	y

square the # and subtract two

x	y

$y = 1/3\,x + 2$

x	y

$y = 3x^2$

x	y

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